Compact Vision–Language Models for Cross-Crop Plant-Disease Diagnosis at the Edge

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Abstract

Cloud vision–language models diagnose plant disease accurately but bill per query and need connectivity; and a conventional classifier has no output unit for a crop absent from its training set. We ask how small a deployable model can be while still diagnosing unseen crops, by matching a leaf image to LLM-written symptom descriptions grounded in cited sources. Using SAGE and frozen compact contrastive language–image encoders: First, frozen encoders from 11.4 to 86.3 million parameters diagnose unseen crops at 2.2–13.9 times chance across nested held-out sets of 16, 34 and 51 classes. Second, which authoring wins depends on scale: a keyword bank leads at 16 and 34 unseen classes (28.9%, 25.0%) but degrades to 21.5% at 51, where source-grounded text reaches 23.9% having risen monotonically. Part of that crossover may be label cleanliness: de-duplicating the largest configuration adds 6.9 points. A matched ungrounded control over seven seeds separates sourcing from per-class distinctness: +1.96 points on the paired classes and +0.94 on the full set, both with intervals including zero. We report this as a null, not as equivalence. Third, a five-item shortlist makes the modest top-1 field-useful: grounded top-5 reaches 60–77% on 16 unseen classes and 59–75% on 51. Fourth, the encoder runs at 17.4 ms per image on a laptop CPU at 45.8 MB; 8-bit quantisation shrinks it to 12.9 MB but costs latency. On known crops the same backbone reaches 82.2% over 166 classes, where supervised CNNs are stronger but cannot transfer at all.

1. Introduction

Cloud VLM diagnosers achieve high in-the-wild accuracy but are metered per query and require connectivity — a poor fit for smallholder and field use, which needs offline, low-cost, real-time inference. A second, deeper obstacle is generalisation: models trained on one crop transfer poorly to others [3], and edge deployment additionally demands compression. Cross-crop transfer is therefore the foundation-model-native open problem this Special Issue targets: it is enabled only by an image–text-aligned VLM, not by a conventional classifier.

We study the question: *how small can a model be and still perform cross-crop disease zero-shot at the edge?* Our system is one **frozen** compact Contrastive Language–Image Pre-training (CLIP) encoder with two heads and an abstain router: a trained head for accurate real-time diagnosis on known crops, and a descriptor head that diagnoses unseen crops by matching image embeddings to large-language-model (LLM) authored, source-grounded symptom-descriptor text. Only the image encoder ships to the device; text prototypes are precomputed offline.

Contributions

1. **Frozen compact VLMs do cross-crop zero-shot.** With source-grounded descriptors they diagnose unseen crops at 2.2–13.9× chance across the 16-, 34- and 51-class held-out sets, taken over the best and worst encoder–configuration cells; averaged over the four tiers the range is 3.4–12.2×. The upper end is partly the denominator falling as the label space grows: against the majority-class prior of Section 4.4 the same result is 1.5–5.7×, which is the comparison we regard as the honest one. On the 17-class pilot survey (Protocol P) accuracy does not track parameter count over a 28-fold range either — model size is not the bottleneck (Sections 4.1–4.3).
2. **A nested scale study isolating what descriptor authoring actually buys.** Symptom text beats a bare class name at every scale (+8.8 points at 16 classes for source-grounded text), and source-grounded descriptors keep improving as the unseen label space triples. We further show that the usual comparison against a keyword-retrieved symptom bank is confounded by prototype collisions rather than by grounding, and separate the two with an ungrounded-generation control (Section 4.3).
3. **Top-5 shortlisting, with a margin-ranked abstain gate** makes the modest top-1 field-useful: the shortlist carries the usefulness and the gate orders confidence so the system can decline (Section 4.4).
4. **A per-tier edge benchmark** with real INT8 sizes and CPU latency, and a quantisation result with practical reach: INT8 helps transformer encoders but *hurts* hybrid convolution–transformer encoders, with a graph-level diagnosis and a deployment rule (Section 4.8).
5. **Rigorous negatives that save effort:** a supervised CNN cannot transfer cross-crop at all, weight-space

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interpolation trades unseen accuracy for seen accuracy rather than giving both, and a leave-one-crop-out analysis shows the held-out crops are *easier* than the trained pool on this split — a limitation we report rather than a control we pass.

2. Related work

Agricultural VLMs SAGE [1] provides our data and the cloud agent we contrast against; its {value, source_url, verbatim_quote} source-grounded schema and its reported 14–16 point gain from symptom knowledge motivate our descriptor design. We are the deployable edge counterpart at $0/image. The wider agricultural foundation-model landscape is surveyed by Li et al. [10], whose taxonomy frames the gap we target: adaptation and deployment cost rather than raw capability. Domain-adapted contrastive models have since appeared — AgriCLIP [14] aligns a CLIP backbone to agriculture and livestock imagery, and SCOLD [15] targets leaf disease specifically — but both are cloud-scale, and we evaluate the latter as a candidate backbone in Section 4.2. Lu and Chen [11] report that zero-shot vision–language models still underperform supervised baselines on plant disease by a wide margin, which is consistent with the modest absolute top-1 we report and is why we lead with shortlisting and abstention rather than raw top-1. Our position differs from the compression literature in kind: we keep an off-the-shelf encoder frozen rather than distilling a student, which removes the student-training step and its accompanying reproducibility burden.

Descriptor-based classification Descriptor-based CLIP classification (DCLIP) [12] and Customized Prompts via Language models (CuPL) [17] classify by LLM-written visual descriptions. In contrast, our descriptors are source-grounded and therefore auditable, compressed to the edge, and evaluated for *cross-crop* agricultural transfer rather than general retrieval.

Compact CLIPs CLIP [19] established image–text contrastive pretraining; MobileCLIP [25], its successor MobileCLIP2 [4] and TinyCLIP [28] compress it, and provide our frozen backbones. We use the open_clip [9] implementations throughout, with SigLIP 2 [23], whose sigmoid objective follows Zhai et al. [29], as a reference ceiling. We contribute an agricultural cross-crop zero-shot study and an edge efficiency analysis rather than a retrieval benchmark.

Robust fine-tuning WiSE-FT [27] explains our forgetting result and motivates the frozen-backbone plus weight-ensembling design.

Domain foundation models We evaluate the biological foundation model BioCLIP 2 [5] and the leaf-disease model SCOLD [15] as candidate backbones and find they do not transfer competitively under our descriptor protocol: the better of the two reaches 1.6× chance against 4.6× for a general-purpose 11.4 M encoder (Section 4.2).

Vision-only foundation models Self-supervised encoders such as DINOv2 [16] learn strong visual features but have no text tower, so they cannot match an image against a written symptom description. Zero-shot diagnosis of an unseen crop therefore requires an image–text-aligned model by construction, which is why every backbone we evaluate is CLIP-family.

Supervised baselines Our comparison set spans ResNet [6], MobileNetV3 [7] and MobileNetV4 [18] among others (Section 4.6).

3. Materials and methods

3.1. Dataset and the nested design

We use SAGE [1] only, drawing images from the released dataset [2] — ∼839 K images over 335 crops and 1,251 disease classes, released under the MIT licence — to avoid the "old, lab-only" critique of legacy corpora such as PlantVillage [13], on which in-distribution accuracy has long been saturated. We stream-filter shards to our crops with content-hash de-duplication, storing images at a longest edge of 288 pixels since every model here operates at 224. Classes are capped at 600 images for held-out crops and 1,000 for seen crops, and a class needs at least 25 images to enter the study at all; images are taken in sorted filename order up to the cap. Filenames in the released dataset are content hashes, so this order is uncorrelated with capture time or any other property of the image: the cap acts as a deterministic random sample rather than favouring, say, the earliest-collected photographs, and the subset is exactly rebuildable from the released code. The caps bound class imbalance rather than save disk: they are what keep the majority-class baseline at 4.2% on the largest held-out configuration, so a frequency prior cannot masquerade as transfer. Nine of the 51 held-out classes reach the 600-image cap, so the held-out set is a capped sample of the pinned revision rather than the whole of it. The resulting subset is 84,123 images over 217 classes and 18 crops: 14,204 images in the 51 held-out classes and 69,919 in the 166 seen classes.

We pin the dataset to revision `bc9bd2899f` (7 May 2026). This is not a formality: SAGE was re-released on 24 August 2026 with re-canonicalised crop and disease names, and the later release drops Cotton entirely — its own canonicalisation record marks every Cotton entry as having no canonical crop. Evaluating against the later release would therefore yield seven held-out crops and 48 classes at our largest configuration rather than the eight and 51 reported here. Every number in this article refers to the pinned revision, and the fetch code pins the same commit hash.

To test whether the approach *scales* rather than reporting a single convenient split, we define three **nested** configurations — every seen and held-out crop in A also appears in B, and B in C — so differences are attributable to the size of the label space and not to which crops were chosen. Configuration A holds out Coffee, Orange and Peach (16 classes); B adds Cotton, Wheat and Bean (34); C adds Banana and Cucumber (51). Seen-set sizes are given in

Table 4. Held-out crops are never used for training at any stage; they are diagnosed only from descriptor text.

3.2. Source-grounded descriptors

Each disease carries a symptom record with fields for pathogen, affected organs and visual symptoms, each field carrying a value, a source URL and a verbatim quote. Records are generated by an LLM under a strict grounding prompt that forbids un-cited claims, then audited. Because the generation endpoint cannot browse, raw citations are model-recalled; we therefore web-verify the headline held-out diseases, replacing citations with sentences copied verbatim from reachable authoritative pages. The symptom text — the string that becomes the CLIP text prototype — is the functional component; the verified citations provide the auditability guarantee. Records that could not be grounded are stored as explicit stubs and are excluded by the loader, so a placeholder can never become a prototype.

3.3. Architecture

One frozen image–text encoder per tier, plus: (1) a trained seen-crop linear probe; (2) a zero-shot descriptor head selecting the nearest source-grounded text prototype by cosine similarity; (3) WiSE-FT weight ensembling, interpolating fine-tuned and frozen visual weights at ratio α; and (4) an abstain router on the top-1 minus top-2 similarity margin. Only the image encoder deploys. We evaluate four deployable tiers (11.4, 21.5, 35.8 and 86.3 M image-encoder parameters) plus a 92.9 M reference ceiling.

3.4. Descriptor strategies and the grounding control

Unless stated otherwise, accuracy is image-weighted (micro-averaged) top-1 over the held-out split, and encoder means are unweighted means over the four deployable tiers, excluding the 92.9 M reference encoder. The class distribution is capped but not balanced, so we also checked the scaling result under crop-macro averaging, which gives 21.9%, 23.8% and 23.9% at configurations A, B and C against the micro values of 21.5%, 22.7% and 23.9%; the monotone rise holds under either convention.

We compare four prompt strategies of increasing specificity, all evaluated with the same frozen encoders and the same held-out splits. *Bare* uses the class name alone. *Crude* adds one generic keyword sentence. *Rich* retrieves a hand-curated symptom paragraph from a 32-key keyword bank. *Grounded* uses the LLM-authored, source-cited symptom text of Section 3.2, falling back to *rich* for any class without a record.

Why rich *cannot isolate grounding* Comparing *grounded* against *rich* conflates two different properties. Retrieval assigns text by matching a disease name against 32 generic keys, so distinct classes routinely receive identical prototypes: at configuration C only 8 of the 51 held-out classes receive a unique entry, 17 fall back to the bare class name, and 26 share text with another class, the largest groups being seven classes keyed on *spot* and five on *blight*. Collisions grow with the label space (2, 14 and 26 classes at A, B and C) while coverage stays flat (68.8%, 61.8%, 66.7%). Any advantage over this baseline therefore measures per-class *distinctness* rather than sourcing. This is a limitation of symptom-bank retrieval as we implemented it, not of per-class generation methods such as DCLIP or CuPL, which emit text per class and cannot collide.

The ungrounded control To isolate the sourcing constraint we generate a matched *ungrounded* arm: the same model, the same schema, the same temperature and the same seeds as *grounded*, with only the requirement to cite a retrievable source removed. Because the two arms differ in exactly one instruction, the difference between them estimates the effect of sourcing alone.

Paired-subset evaluation Coverage differs between the arms, and a naive comparison would absorb that difference. Since `text_for` falls back to the keyword bank wherever an arm has no record, an arm covering 41 of 51 classes is four-fifths its own text and one-fifth shared bank text, which dilutes both arms toward a common baseline. We therefore evaluate both arms on the intersection of classes that every seed of both arms filled genuinely, so that the sourcing requirement is the only difference between them. The intersection is smaller than the full held-out set and chance rises accordingly; we report both. The generation seed is the replication unit. Each seed draws an independent pair of descriptor sets, one per arm, and the same seven seeds are then scored by every encoder on the same images; the four encoders are therefore repeated readouts of one randomised experiment rather than four independent replications of it. We accordingly form the inferential interval over seeds, averaging each seed's difference across encoders first, and report the per-encoder differences separately as corroboration: agreement in sign across four architecturally distinct backbones is evidence that an effect is not an artefact of one encoder, but it is not additional sampling of the randomised quantity and we do not treat it as such.

4. Results

4.1. A frozen 11 M model already does cross-crop zero-shot, and accuracy is flat with size

Across the nested splits (Protocol N) the four deployable encoders span only 19.3–24.6% (rich) and 21.6–27.3% (grounded) at configuration C despite a 7.6-fold spread in parameters, and the same absence of a trend holds over a wider survey: across ten public encoders spanning 11.4–321.8 M on the 17-class pilot set (Protocol P), the largest is only 1.1 points above the smallest despite 28× the parameters, and accuracy does not track size (Spearman $\rho = 0.35$, $p = 0.32$, i.e. not distinguishable from no association; Table 1). That survey includes every publicly available aligned encoder we could load, with no exclusions, so the correlation is over the full set rather than a chosen subset. The 8.5-point spread across that survey is driven by pretraining recipe rather than capacity: the two variants of the same 86 M architecture

Table 1
Pilot encoder survey (Protocol P: 17 held-out classes, chance 5.9%, keyword-bank descriptors). All ten publicly available aligned encoders we could load with open_clip are included; none is excluded. Accuracy does not track parameter count (Spearman $\rho = 0.35$, $p = 0.32$).

Encoder	Params (M)	Zero-shot top-1 (%)
MobileCLIP2-S0	11.4	19.9
MobileCLIP-S1	21.5	20.8
MobileCLIP-S2	35.8	17.7
MobileCLIP2-S2	35.8	20.3
MobileCLIP-B	86.3	21.9
MobileCLIP2-B	86.3	17.1
ViT-B-16-SigLIP2	92.9	25.6
MobileCLIP2-S3	125.2	19.5
MobileCLIP2-L-14	304.0	21.6
MobileCLIP2-S4	321.8	21.0

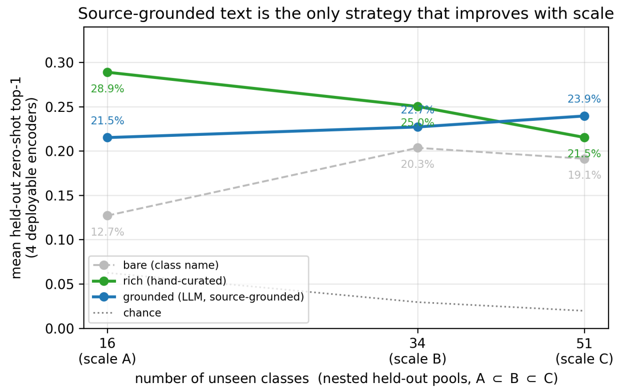

Figure 1: Mean zero-shot accuracy over the four deployable encoders as the unseen label space grows. Source-grounded descriptors improve as the label space widens. The keyword-retrieved bank is shown for reference only: its prototypes collide (Section 4.3), so the gap between the two is not a measurement of grounding.

differ by 4.8 points. Model size is not the bottleneck; the pretrained alignment is the asset. Attempts to *train* a small model to learn this alignment fail — on the earlier 17-class pilot set (Protocol P, not comparable with the nested splits above) a 5.5 M student distilled from scratch reaches only 11.0% against 5.9% chance, versus 19.9% for the frozen 11.4 M encoder it was distilled from; and specialising that encoder on seen crops loses 4.5 points of unseen accuracy to catastrophic forgetting — so we keep the backbone frozen.

4.2. Encoder bake-off

The 11.4 M encoder recovers 86% of the 92.9 M reference at one eighth the parameters, and *beats* the larger 21.5 M model — on the pilot set; on the nested splits of Table 2 the ordering reverses at every configuration, so we read this as recipe noise rather than a ranking. It reinforces that pretraining quality, not size, drives transfer. The one domain model we evaluate faithfully transfers far less well than general-purpose compact encoders under a descriptor protocol: BioCLIP2 reaches 9.6% (1.6× chance) and our best-effort SCOLD load 4.4% (0.75×), against 27.0% (4.6×) for the 11.4 M MobileCLIP2-S0. Taxonomy-style contrastive pretraining does not align to symptom-descriptor text. We note openly that our wrapper for one domain model is a best-effort load and a faithful evaluation would need the authors' inference pipeline; we do not rest any claim on it.

4.3. Descriptor authoring is the lever

Table 2 and Figs. 1 and 2 carry the paper's central result.

Sourced detail is the first lever Detail alone is not enough: a single generic symptom sentence (*crude*) beats a bare class name at configuration A but loses to it at B and C, by 1.6 and 2.4 points. What wins at every scale is *sourced* detail. At the focused scale A, source-grounded text gains 8.8 points on average over a bare class name on frozen 11–86 M edge encoders, and it is the only strategy that improves monotonically as the label space grows.

Source-grounded descriptors keep improving with scale As the unseen label space grows from 16 to 51 classes, source-grounded descriptors improve monotonically (21.5%, 22.7%, 23.9%), and at configuration C they are the strategy that most clearly beats a bare class name. The generated registry covers 156 of 217 classes under a uniform schema.

Sourcing does not measurably cost accuracy The matched ungrounded control of Section 3.4 isolates the effect of requiring a citable source. On the 41 paired classes at configuration C (11,337 images, chance 2.44%), the source-grounded arm leads on all four deployable encoders — +1.76, +2.00, +2.15 and +1.93 points for MobileCLIP2-S0, MobileCLIP-S1, MobileCLIP2-S2 and MobileCLIP-B — for a mean of +1.96 points.

The randomised unit here is the generation seed, not the encoder: the four encoders score the same seven descriptor sets on the same images, so they are four readouts of one experiment rather than four independent replications. Taking seeds as the unit, the per-seed mean differences are +0.69, +1.20, −1.56, +5.66, +1.12, +3.98 and +2.63 points, giving the same +1.96 mean with a 95% confidence interval of $[-0.23, +4.14]$. That interval contains zero. On the full 51-class set the corresponding interval is $[-0.52, +2.39]$, which also contains zero.

We therefore do not claim that source-grounding improves accuracy, and we are careful not to claim the converse either. Failing to reject a null is not evidence of equivalence: with a seed-to-seed standard deviation of 2.4 points and seven seeds, the paired interval spans $[-0.23, +4.14]$ and the full-set interval $[-0.52, +2.39]$, so the largest sourcing penalty these data exclude is about half a point on the full set and rather more on the paired subset. A genuine equivalence claim would need a margin fixed in advance and a two-one-sided test against it; we did not pre-specify one, so we report

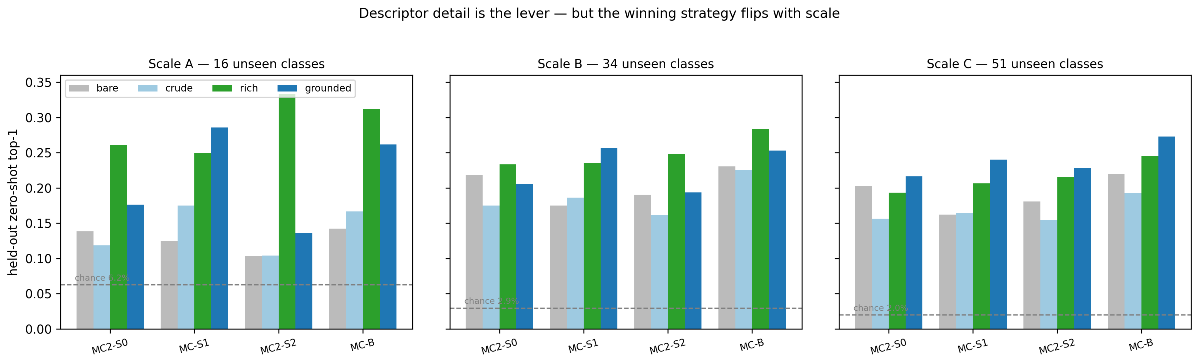

Figure 2: Descriptor strategy by encoder at each held-out scale. Per-encoder top-1 accuracy for the four descriptor strategies at 16, 34 and 51 unseen classes. The source-grounded strategy is the tallest bar for every encoder at the largest scale. The hand-curated bank leads at the smallest scale and still leads on three of the four encoders at the intermediate one, so the crossover is gradual rather than sharp.

the null and the intervals and leave the equivalence question open. At 80% power roughly eleven seeds would resolve an effect of this size, which is the experiment we name as the way to settle it.

The advantage is smaller on the full held-out set The paired subset is the right comparison for isolating the sourcing constraint, but it is not the deployment setting, so we also report the same seven seeds on all 51 held-out classes, where each arm falls back to the keyword bank for the classes it did not fill. The mean advantage halves to +0.94 points and is no longer positive everywhere: +1.89, +0.36, +1.69 and −0.19 points on MobileCLIP2-S0, MobileCLIP-S1, MobileCLIP2-S2 and MobileCLIP-B. The largest encoder reverses. The gap between the two analyses is not dilution but coverage. All ten excluded classes were dropped because the grounded arm failed to produce a record at one or more seeds, while the ungrounded arm filled all 51 at every seed; on those classes the grounded arm therefore falls back to the keyword bank while the ungrounded arm reads its own generated text. The full-set figure is thus the deployment number and it includes a real cost: where grounded generation refuses, its fallback loses to ungrounded generation. That cost is the other face of the over-refusal behaviour described in Section 6, and it is a property of the grounding constraint rather than an artefact of the comparison.

Description length does not explain the difference Because the two arms exceed the encoder's 77-token context, an obvious hypothesis is that the effect depends on how much of each description the encoder reads. Regenerating both arms under a 50-word ceiling, so that neither is truncated, tests it directly. Holding the class set at all 51, capping the descriptions moves the mean from +0.94 to +1.13 points — shortening the text slightly *raises* the advantage rather than removing it. The shrinkage from +1.96 to +1.13 is therefore attributable to the class set rather than to truncation, and we make no claim that the sourcing advantage depends on description length. We return to the token limit as a design constraint in Section 6.

4.4. Top-5 and a margin-ranked abstain gate

The gain grows against the honest baseline Chance is a weak reference for an imbalanced label space, so we also compare against the majority-class frequency prior. At configuration A that prior is 14.8% against grounded's 21.5%, a modest margin; at configuration C it falls to 4.2% while grounded rises to 23.9%. The head therefore becomes *relatively* stronger as the label space widens, which is the opposite of the usual pattern and the reason we report accuracy at three nested scales rather than one.

A field tool can propose a short list and abstain when unsure, so raw top-1 is not the operative metric. Table 3 and Fig. 3 show grounded top-5 at 60–77% on 16 classes, holding at 59–75% on 51 classes, against a 9.8% five-item random shortlist at that scale. Selective accuracy rises monotonically as coverage tightens, confirming that the margin orders confidence correctly, but the gate's effect size is small: refusing the least confident fifth of images buys 2.4, 2.1 and 2.0 points of top-1 at configurations A, B and C. The shortlist, not the gate, is what makes the system field-useful; the gate's value is that the system can decline rather than guess. Selective accuracy rising with coverage, confirming the confidence signal is correctly ordered rather than inverted — a property that holds for every encoder, every strategy and every scale we measured. The grounded strategy's abstention advantage, by contrast, is scale-dependent: at the largest label space it holds the highest top-5 on all four deployable encoders and the lowest area under the risk–coverage curve on three of them, whereas at the smallest label space it leads on neither sweep. We therefore recommend it as the deployment setting for large label spaces, which is the regime the deployment case actually concerns, and note that at 16 classes the hand-curated bank remains competitive.

4.5. Known crops, and the seen/unseen dial

The same frozen backbone with a linear probe reaches 82.2% over 166 fine-grained classes (Table 4, Fig. 4), against 9.3% for the same encoder used zero-shot on its own seen classes. Two patterns repeat here. Accuracy is flat across the same 7.6-fold parameter range (≤1.5 points at every scale),

Table 2
Cross-crop zero-shot accuracy at three held-out scales, by descriptor strategy.

Model	Params (M)	bare	crude	rich	grounded
Config A — 16 unseen classes, chance 6.2%					
MobileCLIP2-S0	11.4	13.8%	11.9%	26.1%	17.6%
MobileCLIP-S1	21.5	12.4%	17.5%	24.9%	28.6%
MobileCLIP2-S2	35.8	10.3%	10.4%	33.3%	13.7%
MobileCLIP-B	86.3	14.2%	16.7%	31.2%	26.2%
ViT-B-16-SigLIP2	92.9	17.1%	19.6%	30.4%	26.8%
mean	—	**12.7%**	**14.1%**	**28.9%**	**21.5%**
Config B — 34 unseen classes, chance 2.9%					
MobileCLIP2-S0	11.4	21.8%	17.5%	23.3%	20.5%
MobileCLIP-S1	21.5	17.5%	18.6%	23.6%	25.6%
MobileCLIP2-S2	35.8	19.0%	16.1%	24.8%	19.4%
MobileCLIP-B	86.3	23.1%	22.6%	28.4%	25.3%
ViT-B-16-SigLIP2	92.9	27.1%	26.2%	29.9%	27.3%
mean	—	**20.3%**	**18.7%**	**25.0%**	**22.7%**
Config C — 51 unseen classes, chance 2.0%					
MobileCLIP2-S0	11.4	20.2%	15.6%	19.3%	21.6%
MobileCLIP-S1	21.5	16.2%	16.4%	20.7%	24.0%
MobileCLIP2-S2	35.8	18.1%	15.4%	21.5%	22.8%
MobileCLIP-B	86.3	22.0%	19.3%	24.6%	27.3%
ViT-B-16-SigLIP2	92.9	22.0%	20.8%	23.1%	28.4%
mean	—	**19.1%**	**16.7%**	**21.5%**	**23.9%**

bare = class name only; *rich* = hand-curated symptom paragraph; *grounded* = LLM source-grounded symptom text. Held-out crops are never trained on. Crop pools are nested (A ⊂ B ⊂ C), so differences are attributable to the size of the unseen label space. Note the reversal: mean hand-curated accuracy falls as classes grow while source-grounded accuracy rises.

Table 3
Top-5 and selective prediction, source-grounded against the hand-curated bank. acc@cov80 = accuracy when the 80% most confident predictions are kept.

			grounded				rich			
Config	Classes	Model	Top-1	Top-5	AURC ↓	acc@cov80	Top-1	Top-5	AURC ↓	acc@cov80
A	16	MobileCLIP2-S0	17.6%	61.9%	0.6986	19.4%	26.1%	59.6%	0.6776	28.9%
A	16	MobileCLIP-S1	28.6%	60.3%	0.5015	33.0%	24.9%	55.3%	0.6287	28.6%
A	16	MobileCLIP2-S2	13.7%	61.2%	0.7426	14.8%	33.3%	63.6%	0.584	37.6%
A	16	MobileCLIP-B	26.2%	77.1%	0.6141	28.4%	31.2%	69.9%	0.5243	35.0%
A	16	ViT-B-16-SigLIP2	26.8%	76.9%	0.6532	28.5%	30.4%	68.5%	0.6197	33.4%
B	34	MobileCLIP2-S0	20.5%	70.0%	0.7077	22.3%	23.3%	63.3%	0.6775	25.1%
B	34	MobileCLIP-S1	25.6%	63.7%	0.5944	29.2%	23.6%	61.5%	0.6018	26.6%
B	34	MobileCLIP2-S2	19.4%	71.6%	0.7128	20.3%	24.8%	66.4%	0.6114	27.1%
B	34	MobileCLIP-B	25.3%	76.0%	0.5941	27.4%	28.3%	67.5%	0.545	30.5%
B	34	ViT-B-16-SigLIP2	27.3%	74.7%	0.6623	28.3%	29.9%	72.4%	0.5721	32.4%
C	51	MobileCLIP2-S0	21.6%	62.9%	0.7168	23.7%	19.3%	61.7%	0.7748	20.2%
C	51	MobileCLIP-S1	24.0%	59.0%	0.643	26.9%	20.7%	58.1%	0.7008	22.6%
C	51	MobileCLIP2-S2	22.8%	68.3%	0.7356	23.6%	21.5%	66.1%	0.697	23.4%
C	51	MobileCLIP-B	27.3%	74.6%	0.6169	29.7%	24.6%	66.0%	0.6548	25.8%
C	51	ViT-B-16-SigLIP2	28.4%	67.7%	0.6155	30.7%	23.1%	67.1%	0.6813	24.9%

Confidence is the top-1 minus top-2 similarity margin. Selective accuracy rising as coverage tightens confirms the signal is correctly ordered; the gate itself buys about two points of top-1 for refusing one image in five.

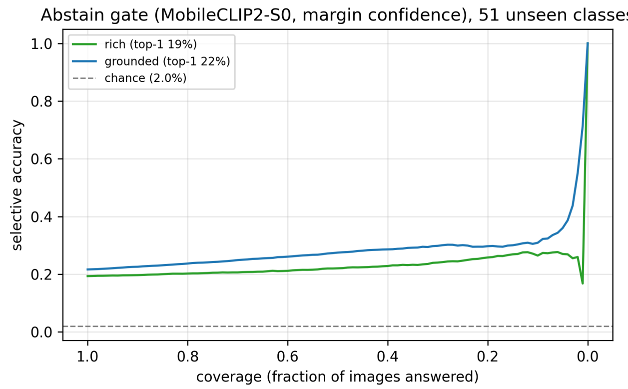

Figure 3: Risk–coverage behaviour of the abstain gate. Selective accuracy as a function of the fraction of images the model chooses to answer, using the top-1 minus top-2 similarity margin as the confidence score. Accuracy rises monotonically as coverage tightens, confirming the score is correctly ordered.

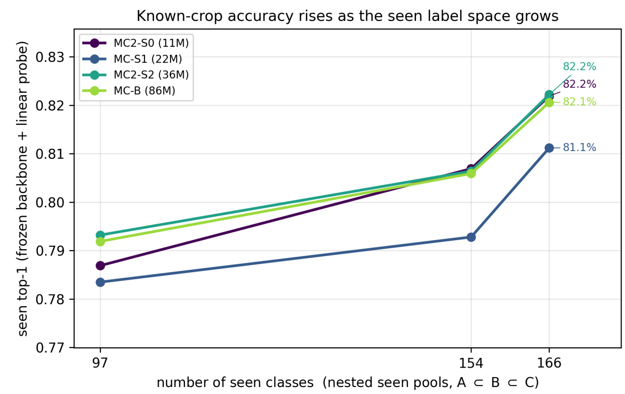

Figure 4: Known-crop accuracy against the size of the seen label space. Linear-probe top-1 for four frozen encoders at 97, 154 and 166 seen classes. Accuracy rises with the label space and is flat across a 7.6-fold parameter range.

mirroring Section 4.1; and accuracy *rises* as the seen label space grows (by roughly three points from 97 to 166 classes, for every encoder), the opposite of the unseen side, because additional crops contribute proportionally more training images than decision difficulty. Each head is therefore deployed where its scaling behaviour is favourable. WiSE-FT (Table 5, Fig. 5) exposes the trade-off as a single interpolation weight, though it is not a clean monotone dial. Interpolation is not free at the midpoint: $\alpha = 0.5$ buys only 1.3 points of seen accuracy while costing 6.2 points of unseen. It pays off at the upper end, where $\alpha = 0.75$ gains 8.8 points of seen for 3.2 of unseen and full fine-tuning ($\alpha = 1$) gains 22.5 for 5.2. Neither column is monotone in α — $\alpha = 0.75$ beats $\alpha = 0.25$ on both — so the weight selects an operating point rather than sliding along a frontier. One caveat is large enough to bound the whole reading. This sweep subsamples the seen split to 200 images per class, and its own frozen reference is 58.8% against the 82.2% that Table 4 reports for the same encoder on the full split. Full fine-tuning reaches 81.5%, still slightly below that full-split frozen probe, so the 22.5 points the dial appears to buy may be recovering the subsample deficit rather than adding seen accuracy. We report the sweep as evidence that the interpolation behaves, not as a measurement of what fine-tuning is worth; settling that needs the sweep re-run on the full split. The unseen side is unaffected by the subsampling, and there naive fine-tuning does not destroy cross-crop transfer: unseen accuracy settles at 16.4%, still 8.4 times chance, so the dial trades transfer for seen accuracy rather than erasing it.

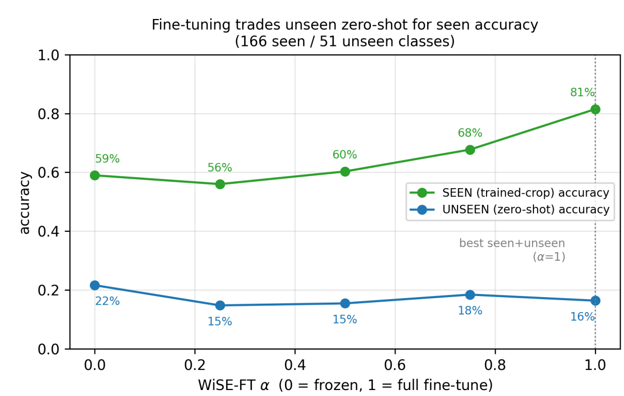

Figure 5: The seen–unseen trade-off as a single interpolation weight, on the same sweep as Table 5. Accuracy on seen and unseen crops as the weight-space ensembling ratio α moves from the frozen encoder ($\alpha = 0$) to the fully fine-tuned one ($\alpha = 1$). Seen accuracy rises by 22.5 points while unseen accuracy falls by 5.2, from 21.6% to 16.4%; transfer is traded down rather than destroyed, since 16.4% remains 8.4× the 1.96% chance level. The seen column carries a few points of head-refit noise, so the trend rather than any single interior point is the readable quantity.

4.6. Supervised baselines and leave-one-crop-out

Fourteen supervised baselines — twelve convolutional networks and two hybrid convolution–transformer models, drawn from the ResNet [6], DenseNet [8], MobileNetV3 [7], MobileNetV4 [18], EfficientNet [21] and EfficientNetV2 [22], RegNetY [20], ConvNeXt V2 [26] and FastViT [24] families — spanning 1.7–42.8 M parameters

Table 4
Known-crop accuracy: frozen backbone plus a linear probe, at three seen-set sizes.

Config	Crops	Classes	Images	MobileCLIP2-S0	MobileCLIP-S1	MobileCLIP2-S2	MobileCLIP-B
A	4	97	42,326	78.7%	78.3%	79.3%	79.2%
B	8	154	62,043	80.7%	79.3%	80.6%	80.6%
C	10	166	69,919	82.2%	81.1%	82.2%	82.1%

The same frozen encoders that perform unseen-crop zero-shot (Table 2). Accuracy rises with the seen label space because additional crops contribute proportionally more training images than difficulty.

Table 5
WiSE-FT weight ensembling on MobileCLIP2-S0: the seen/unseen trade-off as a single dial (166 seen classes, 23,445 images; 51 unseen classes; protocol: nested-C).

α	Seen ↑	Unseen zero-shot ↑
0.0 (frozen)	59.0%	21.6%
0.25	56.0%	14.8%
0.50	60.3%	15.5%
0.75	67.7%	18.4%
1.0 (naive fine-tune)	81.5%	16.4%

$\alpha = 0$ reproduces this sweep's own frozen probe (58.8%), validating the interpolation. That probe is measured on the 200-image-per-class subsample used here, not on the full seen split of Table 4. $\alpha = 1$ is naive fine-tuning; here it still retains 8.4× chance on unseen crops, so the dial trades cross-crop transfer for seen accuracy rather than destroying it. $\alpha = 1.00$ maximises the sum of the two columns.

are trained on the identical 166-class seen set under one protocol: ImageNet-pretrained initialisation, AdamW at a learning rate of 3×10^{-4} with weight decay 10^{-4}, batch size 96, 224 px inputs with random horizontal flips, and 4 epochs (Table 6; per-epoch curves in Fig. 6). The budget is fixed at 4 epochs by session limits rather than by convergence, and this is a real constraint on what the sweep can support. Eleven of the fourteen runs are still improving at the fourth epoch, by a mean of 1.1 points and by as much as 3.4, and the larger models start lower and climb longer — ResNet-101 moves from 77.9% to 86.7% over the four epochs. A longer budget would therefore be expected to favour the larger models, so the size-versus-accuracy comparison below should be read as a statement about this budget rather than about converged performance. They are strong on known crops: the best reaches 89.3%, about 7.1 points above the frozen probe. We report this plainly, because for a fixed, known label set a conventional classifier is the better choice.

Capacity is not the lever, and the complete sweep states it plainly: the strongest of the fourteen is EfficientNet-B0 at 4.2 M with 89.3%, which is ten times smaller than the largest model tested and beats it by 2.5 points — ResNet-101, at 42.8 M, reaches only 86.7%. Across a 25× parameter range the family spans 9.2 points with no monotone trend, and the rank correlation between size and accuracy is weak (Spearman $\rho = 0.36$). That reproduces Section 4.1 and the seen probe above in a third, fully supervised setting, making "size is not the bottleneck" a property of the task rather than an artefact of frozen backbones. We do not claim capacity is irrelevant: the smallest network, MobileNetV3-Small at 1.7 M, is 9.2 points behind the best, so there is a floor. The claim is narrower and better supported — past a few million parameters, additional capacity buys little on this task. The sharpest control is FastViT-SA12 (10.7 M), the same architecture family as our image encoder (11.4 M) at nearly the same size: 86.5% trained supervised against 82.2% as a frozen probe, so the 4.3-point difference is training regime, not architecture. Yet none of the fourteen has an output unit for an unseen class, so cross-crop accuracy is not merely low — it is structurally undefined at any parameter count.

Table 7 shows the two hardest crops overall are both *trained* crops, so difficulty is not simply a function of whether a crop was held out. The held-out crops do, however, average above the trained ones on this split, and we do not claim the held-out set is adversarially hard; the leave-one-crop-out spread (from 4.8% to 28.9% across crops) is better read as evidence that per-crop difficulty varies far more than the held/trained distinction does.

4.7. On-device efficiency

Table 8 and Fig. 7 report the deployable image encoder on a laptop processor. The 11.4 M tier runs at 17.4 ms/image (~58 img/s), with ONNX Runtime about 2.7× faster than eager PyTorch, and compresses 3.5× to 12.9 MB under INT8.

4.8. INT8 is architecture-dependent, not universally beneficial

A naive dynamic-quantisation call — the default recipe in most tutorials — makes the lightweight tiers 18–19× *slower*. A properly configured static quantise–dequantise (QDQ) pipeline (shape-inference pre-pass, per-channel weights, calibrated activations) recovers 5–9× of that but still does not beat FP32 on those tiers. The pure-transformer model behaves oppositely and becomes 2.2× faster.

The mechanism is visible in the quantised graphs: counting convolutions the quantiser could not convert gives 119, 215 and 235 for the three hybrid tiers against 3 for the pure transformer. On the hybrids the runtime must dequantise, run a float convolution, and requantise hundreds of times per

Table 6
Supervised CNN baselines on the identical seen set.

Architecture	Params (M)	Classes	Seen top-1 ↑	Unseen
efficientnet-b0	4.2	166	89.3%	**n/a (structural)**
tf-efficientnetv2-s	20.4	166	88.7%	**n/a (structural)**
mobilenetv3-large-100	4.4	166	87.9%	**n/a (structural)**
densenet121	7.1	166	87.6%	**n/a (structural)**
resnet50	23.9	166	87.6%	**n/a (structural)**
convnextv2-tiny	28.0	166	87.1%	**n/a (structural)**
convnextv2-nano	15.1	166	87.0%	**n/a (structural)**
regnety-040	19.7	166	86.9%	**n/a (structural)**
resnet101	42.8	166	86.7%	**n/a (structural)**
fastvit-sa12	10.7	166	86.5%	**n/a (structural)**
mobilenetv4-conv-medium	8.7	166	84.9%	**n/a (structural)**
mobilenetv4-conv-small	2.7	166	83.2%	**n/a (structural)**
fastvit-t8	3.4	166	82.5%	**n/a (structural)**
mobilenetv3-small-100	1.7	166	80.0%	**n/a (structural)**

For a fixed, known label set a CNN is the stronger classifier. Accuracy does not track parameter count: the strongest network is efficientnet-b0 at 4.2 M, which is 10× smaller than the largest model here and beats it by 2.5 points. None of them, however, has an output unit for an unseen class, so cross-crop accuracy is not low but undefined — the capability the descriptor head supplies.

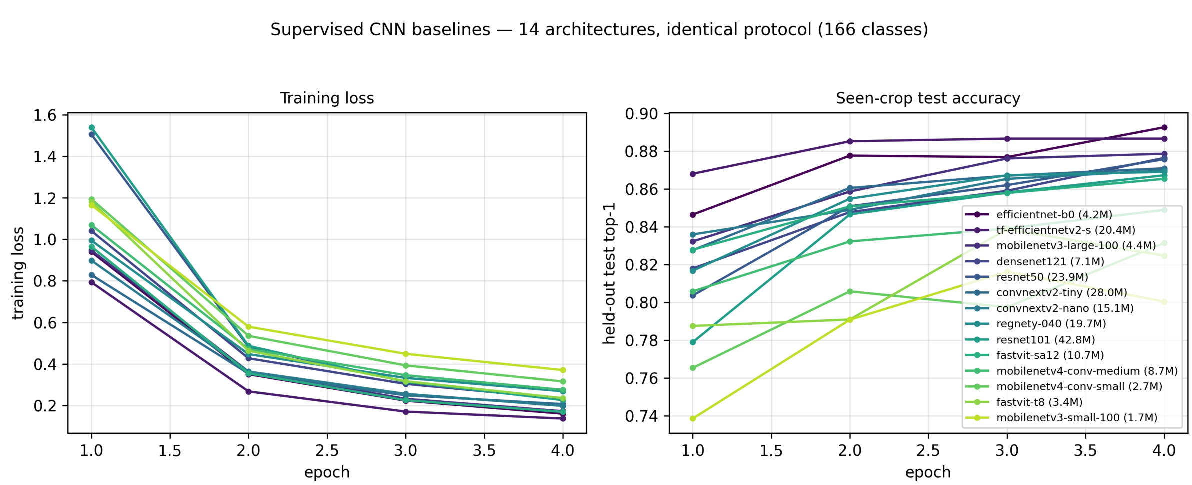

Figure 6: Training behaviour of the fourteen supervised baselines. Training loss and held-out test accuracy per epoch under one protocol. Eleven of the fourteen are still improving at the fourth epoch, so the budget bounds the comparison rather than representing converged performance; the smallest models show the largest epoch-to-epoch variance.

Table 7
Leave-one-crop-out on MobileCLIP2-S0/dfndr2b (78 classes, chance 1.3%).

Crop	N	Zero-shot ↑	95% CI
Orange	1,361	28.9%	[26.4%, 31.3%]
Coffee	1,582	15.8%	[14.0%, 17.7%]
Apple	10,000	12.7%	[12.0%, 13.3%]
Peach	1,107	12.3%	[10.4%, 14.2%]
Potato	2,698	9.5%	[8.3%, 10.7%]
Corn	9,403	4.8%	[4.4%, 5.3%]
Pooled	**26,151**	**10.5%**	[10.2%, 10.9%]

The two hardest crops are both *trained* crops, so difficulty does not follow the held-out boundary. The held-out crops nevertheless average above the trained pool here (19.0% against 9.0%), so this split is not adversarially hard. Measured with the `rich` strategy on the 78-class leave-one-crop-out pool, not the headline label space.

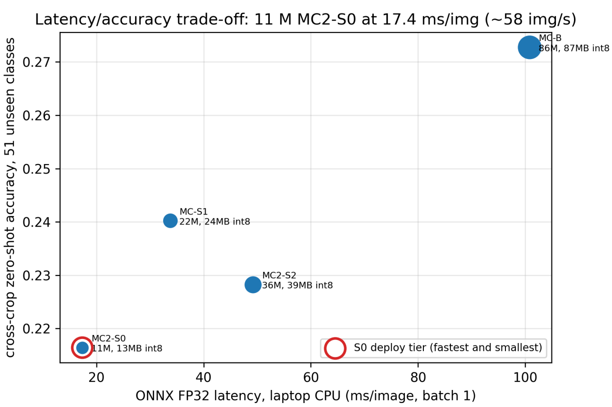

Figure 7: Accuracy against on-device latency. Cross-crop zero-shot accuracy versus single-image latency on a laptop processor; marker area is proportional to parameter count and labels give the 8-bit footprint. The 11.4-million-parameter tier is the deployment sweet spot.

inference; dynamic quantisation instead maps every convolution to an integer-convolution operator that is pathologically slow for depthwise kernels on x86.

Practical guidance For hybrid convolution–transformer encoders on a CPU runtime, INT8 is a *size* lever, not a *speed* lever; deploy FP32. For transformer encoders it is both. We flag rather than assume that Arm processors, whose vector extension provides well-optimised INT8 depthwise kernels, may reverse this for the hybrid tiers.

Table 8
On-device cost of the deployable image encoder (CPU, batch 1, 224×224). The four latency columns are milliseconds; the final two are megabytes.

Model	Params (M)	Torch FP32 ↓	ONNX FP32 ↓	INT8 dyn. ↓	INT8 static ↓	FP32 ↓	INT8 ↓
MobileCLIP2-S0	11.4	47.1	17.4	320.0	62.2	45.81	12.92
MobileCLIP-S1	21.5	99.8	33.7	614.9	79.5	86.51	24.29
MobileCLIP2-S2	35.8	119.0	49.2	923.8	98.7	143.62	39.16
MobileCLIP-B	86.3	130.7	100.8	60.4	46.4	345.55	87.4

Latency in ms (median of 50 runs), ONNX Runtime 1.26.0. INT8 makes the hybrid conv–transformer tiers *slower* while shrinking them $\sim 3.5\times$; only the pure-transformer model gets faster.

5. Discussion

The contribution is a system, not a backbone The tiers are off-the-shelf encoders used frozen; the novelty is the source-grounded descriptor head, the margin-ranked abstain gate, the WiSE-FT knob, and the real-time edge packaging. Because accuracy is flat with size, the reason to offer a family is the measured latency/size Pareto, not accuracy.

Source-grounding is a trust contribution, and the authoring route that scales Grounded generation requires one retrievable source per class, where hand-curation requires an expert per crop and, in the retrieval form used here, cannot even keep prototypes distinct as the label space grows. That asymmetry is what we consider the paper's transferable claim beyond agriculture, and the ungrounded control in Section 4.3 settles the question it raises: the sourcing constraint does not measurably cost accuracy. The point estimate is positive on every encoder and at both class sets — +1.96 points on the paired classes and +0.94 on the full held-out set — but the seed-level intervals include zero, so we claim the absence of a penalty rather than a gain. The practical reading is not that grounding is an accuracy technique — two points would be a thin reason to adopt anything — but that the trade-off practitioners might expect to face between auditability and performance does not appear in our measurements. Where accuracy *does* move is descriptor authoring as a whole: at the focused scale, replacing a class name with a full symptom description is worth 8.8 points, four times the sourcing effect. Authoring is the lever; sourcing is what makes the lever safe to pull in a domain where a wrong answer costs a harvest.

Honesty on absolutes Fine-grained cross-crop top-1 is modest (14–29%); we lead with top-5 and the margin-ranked abstain gate as the field-relevant metrics, and frame cross-crop transfer as a hard open problem we advance at edge scale rather than claim to solve.

6. Limitations

Descriptor coverage and two tiers of citation Across 18 crops, 156 of 217 disease records are LLM-filled; the remaining 61 are explicit stubs that fall back to the hand-curated bank. Of the 156, 112 carry a verbatim quote. We distinguish two tiers of provenance. A quote is *page-verified* when we retrieved the cited page and read the sentence there; it is *model-recalled* when the generating model supplied both quote and URL without browsing. Forty records carry at least one page-verified field, and in 16 of those every cited field is page-verified. Verification was prioritised by where it matters: of the 47 filled records on held-out crops — the set carrying this paper's cross-crop claim — 23 are verified. Model-recalled quotes are reported as unverified provenance rather than as evidence, and the per-URL audit trail is released with the data so any citation can be checked independently.

The grounding prompt over-refuses Because the generation endpoint cannot browse, a strict "never cite what you cannot quote" instruction makes the model return empty records rather than risk fabrication, including for well-documented diseases. A bulk re-run converted only 1 of the 61 stubs. This is the anti-hallucination guarantee behaving correctly, but it means grounded coverage scales with human-supplied sources, not with more API spend.

Label noise in the dataset Several held-out "diseases" are not diseases: three are breeding resistance ratings, for which no symptom descriptor can exist; one is an insect pest rather than a pathogen; and one is a misspelling of a genus. A second defect is duplication: five pairs of held-out labels name the same disease twice — `Orange/Canker` and `Orange/Citrus_Canker`, `Orange/Greening_Disease` and `Orange/Huanglongbing`, `Peach/Leaf_Curl` and `Peach/Peach_Leaf_Curl`, `Cucumber/Angular_Leaf_Spot` and `Cucumber/Angular_Leaf_Spot_Of_Cucumber`, and `Wheat/Head_Scab` and `Wheat/Fusarium_Graminearum_Schwabe`. No descriptor can separate a synonymous pair, so each such class carries an accuracy ceiling near 50%.

This bears on the scaling result specifically, and we state the risk rather than leave it to be found. Duplicate density falls as the label space grows — 6 of 16 evaluated classes at configuration A, 8 of 34 at B and 10 of 51 at C, or 37.5%, 23.5% and 19.6% — which is the same direction as the rise in grounded accuracy. A sceptical reading of Section 4.3 is therefore available: the label space may be getting cleaner rather than the method getting relatively better, and the runs reported here cannot fully separate the two. What we can report is a de-duplicated evaluation at configuration C, released as `zeroshot_eval_C_clean.json`: removing one label from each of the five alias pairs, and four of the five non-disease labels, leaves 42 classes and raises the grounded mean from 23.9% to 30.8%. Cleaning helps the method, so reporting the uncleaned numbers as the headline is conservative, but the equivalent runs at A and B do not yet exist and the scaling claim should be read as provisional until they do. We recommend excluding both the rating labels and the alias pairs in future use of this benchmark.

Descriptor length exceeds the text context The encoders inherit CLIP's 77-token text window, and our generated descriptions are longer than it: essentially every generated prototype is truncated, and the ungrounded arm, whose descriptions are the longer of the two, loses proportionally more of its text. The truncated portions retain visual symptom vocabulary in almost every case, so the comparison remains meaningful, but it is a comparison between the leading sentences of each description rather than between the records as authored. A descriptor protocol written to fit the context window — visual symptoms first, taxonomy last, or a summarisation pass before embedding — is the obvious next refinement, and we expect it to raise both generated arms rather than to separate them.

Scope and variance Configuration C covers 18 of the dataset's crops, not all of it. The zero-shot, probe, abstention and supervised results are single-seed on a single evaluation machine, and we report no variance estimates for them. The control arm of Section 4.3 is the exception, and it is the analysis our central claim rests on: it runs seven generation seeds per arm and we report the seed-level interval there. The leave-one-crop-out analysis carries bootstrap intervals over images. A configuration-C re-measurement on an independently rebuilt embedding cache three weeks later reproduced each encoder to within 0.6 percentage points; both records are released rather than summarised by a single agreement figure.

Sub-10 M gap Compact aligned models below about 11 M do exist — TinyCLIP [28] ships an 8 M image tower — but none we could load reached usable cross-crop accuracy under a descriptor protocol, and we did not evaluate them systematically; a ∼5 M tier would require weight-inherited distillation and is framed as future work.

7. Conclusion

A single frozen, compact, descriptor-driven VLM brings cross-crop plant-disease diagnosis to laptops and small devices at $0/image: cross-crop zero-shot at 2.2–13.9× chance across the four deployable tiers, on 16–51 unseen classes, 82.2% on 166 known classes (89.3% for the strongest supervised CNN), grounded top-5 of 60–77% at 16 unseen classes and 59–75% at 51, a margin-ranked abstain gate, and 17.4 ms real-time CPU inference at 45.8 MB, compressible to 12.9 MB under INT8 if size outweighs latency. Model size is a near-non-factor across the 11.4–321.8 M range surveyed on the pilot set, and across the 11.4–86.3 M deployable family in every other experiment. The levers are, in order: the descriptor authoring method — and specifically the finding that source-grounded descriptors keep improving as the unseen label space grows — honest abstention, and the frozen-backbone hybrid.

CRediT authorship contribution statement

Pinni Venkata Abhiram: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, Visualization, Project administration. **Gaurav Shrivastava:** Methodology, Software, Validation, Investigation, Writing – review and editing. **Rahul Ananthasayanam:** Validation, Investigation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data underlying this study are public. The images are the SAGE crop-disease dataset [2] at revision `bc9bd2899f`, released under the MIT licence and used without modification other than the crop and per-class subsetting and the 288-pixel resizing described in Section 3.1. The revision is quoted because a later release of SAGE changes the class inventory; see Section 3.1. The source-grounded descriptor registry, every result file including per-epoch training curves, the table and figure generators, and the full experimental code are released at `https://doi.org/PENDING-ZENODO-DOI`, which reproduces every number and figure in this article from the released result files.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Two distinct uses are declared. First, as a methodological component: a large language model generates the source-grounded symptom descriptions that form the descriptor registry, described in full in Section 3.2. This is an object of study rather than a writing aid, and the generated text is released with the data so that it can be inspected. Second, in preparing the manuscript: the authors used a large language model to assist with drafting and editing prose and with generating the analysis scripts that produce the tables and figures from the released result files. No text was accepted without review, every reported number is regenerated from a released JSON rather than transcribed, and the authors reviewed and edited all content and take full responsibility for the publication.

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